

BRAZIL E-COMMERCE

Exploratory Data Analysis Report

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Summary

This report presents a comprehensive analysis of the Brazilian e-commerce market using the Olist public dataset. The initial analysis (EDA) was conducted through SQL queries by establishing a connection in Jupyter notebook and Power BI and Power Query has been used to effectively visualise the data and present in the format of a dashboard to give recommendations on the direction of the business.

Establishing SQLite connection for EDA

To conduct exploratory data analysis, the tool of choice has been SQL using Jupyter notebook. Jupyter notebook allows for the creation of an engine that can store all the .csv files and establish SQLite connection, as seen the code piece below.

```
import pandas as pd
import os
from sqlalchemy import create_engine

#Create an engine for SQLite
engine = create_engine('sqlite:///ecomm.db')

#File names
for file in os.listdir('data'):
    print(file)

olist_customers_dataset.csv
olist_geolocation_dataset.csv
olist_orders_dataset.csv
olist_order_items_dataset.csv
olist_order_payments_dataset.csv
olist_order_reviews_dataset.csv
olist_products_dataset.csv
olist_sellers_dataset.csv
product_category_name_translation.csv
```

Furthermore, the data has been read from a data frame into SQL tables for the analysis using the below process.

```
for file in os.listdir('data'):
    if '.csv' in file:
        df = pd.read_csv('data/' + file)
        print(df.shape)

(99441, 5)
(1000163, 5)
(99441, 8)
(112650, 7)
(103886, 5)
(99224, 7)
(32951, 9)
(3095, 4)
(71, 2)

# to load data from pandas df to SQL table
def ingest_db(df, table_name, engine):
    df.to_sql(table_name, con = engine, if_exists = 'replace', index = False)

for file in os.listdir('data'):
    if '.csv' in file:
        df = pd.read_csv('data/' + file)
        print(df.shape)
        ingest_db(df, file[:-4], engine)
```

Exploratory Data Analysis

After establishing the connection, six main insights come forth from the queries formed that include payment types, number of orders, freight value, revenue and customer segmentation. The initial process uses the following queries to read data in the SQL table for easier reference.

```
import pandas as pd
import sqlite3

#connect to database connection
connection = sqlite3.connect('ecomm.db')

tables = pd.read_sql_query("SELECT name from sqlite_master WHERE type='table'", connection)
tables
```

```
customers = pd.read_sql("select * from olist_customers_dataset", connection)
customers
geolocation = pd.read_sql("select * from olist_geolocation_dataset", connection)
geolocation
orders = pd.read_sql("select * from olist_orders_dataset", connection)
orders
items = pd.read_sql("select * from olist_order_items_dataset", connection)
items
payments = pd.read_sql("select * from olist_order_payments_dataset", connection)
payments
reviews = pd.read_sql("select * from olist_order_reviews_dataset", connection)
reviews
products = pd.read_sql("select * from olist_products_dataset", connection)
products
sellers = pd.read_sql("select * from olist_sellers_dataset", connection)
sellers
```

- Key Insights from the EDA

1. Top payment types by payment value and number of orders

The following table shows total transaction value and order volume broken down by payment type:

```
#Top payment types by payment value and number of orders
query1 = pd.read_sql("""select p.payment_type, sum(p.payment_value) as total_amount, count(o.order_id) as total_orders
from olist_order_payments_dataset as p
join olist_orders_dataset as o on p.order_id = o.order_id group by p.payment_type""", connection)
query1
```

Payment Type	Total Amount (BRL)	Total Orders
Credit Card	12,542,084.19	76,795
Boleto	2,869,361.27	19,784
Voucher	379,436.87	5,775
Debit Card	217,989.79	1,529

- Credit card is the dominant payment method, accounting for 73.4% of all orders.
- Boleto (a Brazilian bank slip payment method) represents a significant share, indicating a meaningful segment of customers who lack credit card access or prefer other methods of payment.

- Vouchers being used suggest an active promotional or loyalty discount programme.
- Debit card usage is minimal at 1.5%, indicating limited consumer usage of this method.

2. Products with highest average rating and number of orders.

```
# Number of orders in each product category along with average rating
query2 = pd.read_sql("""
    SELECT
        p.product_category_name,
        COUNT(o.order_id) AS total_orders,
        round(avg(r.review_score),2) AS average_rating
    FROM olist_orders_dataset AS o
    JOIN olist_order_reviews_dataset AS r ON r.order_id = o.order_id
    JOIN olist_order_items_dataset AS oi ON oi.order_id = o.order_id
    JOIN olist_products_dataset AS p ON p.product_id = oi.product_id
    WHERE product_category_name != 'None'
    GROUP BY p.product_category_name
    ORDER BY avg(r.review_score) DESC
    limit 10
""", connection)
query2
```

	product_category_name	total_orders	average_rating
0	cds_dvds_musicais	14	4.64
1	fashion_roupa_infanto_juvenil	8	4.50
2	livros_interesse_geral	549	4.45
3	construcao_ferramentas_ferramentas	99	4.44
4	flores	31	4.42
5	livros_importados	60	4.40
6	livros_tecnicos	266	4.37
7	alimentos_bebidas	279	4.32
8	malas_acessorios	1088	4.32
9	portateis_casa_forno_e_cafe	76	4.30

3. Top States by Order Volume

The analysis of the top 5 states by order count reveals a pronounced geographic concentration:

```
#States with highest number of orders
query3 = pd.read_sql("""select c.customer_state, count(o.order_id) as total_orders
from olist_orders_dataset as o
join olist_customers_dataset as c on c.customer_id = o.customer_id
group by c.customer_state
order by count(o.order_id) DESC
limit 5""", connection)
query3
```

Rank	State	Total Orders
1	São Paulo (SP)	41,746
2	Rio de Janeiro (RJ)	12,852
3	Minas Gerais (MG)	11,635
4	Rio Grande do Sul (RS)	5,466
5	Paraná (PR)	5,045

- São Paulo alone accounts for around 41% of all orders, reflecting both its and its role as the country's economic hub.
- The top 3 states (SP, RJ, MG) together indicate that the majority of e-commerce demand is concentrated in the South-East region of Brazil.
- Southern states like RS and PR show moderate activity, while Northern and North-Eastern states have growth potential.

4. Freight Costs by Category (Top 5)

Freight cost is a critical operational expense for the seller of the goods. The top 5 categories by total freight paid are:

```
#Freight value according to product category
query4 = pd.read_sql("""select p.product_category_name, sum(i.freight_value) as total_freight
from olist_products_dataset as p
join olist_order_items_dataset as i on p.product_id = i.product_id group by p.product_category_name
order by sum(i.freight_value) DESC
limit 5""", connection)
query4
```

Rank	Category	Total Freight (BRL)
1	Bed, Bath & Table (cama_mesa_banho)	204,693.04
2	Beauty & Health (beleza_saude)	182,566.73
3	Furniture & Decor (moveis_decoracao)	172,749.30
4	Sports & Leisure (esporte_lazer)	168,607.51
5	Computers & Accessories (informatica_acessorios)	147,318.08

- Beauty & Health leads in freight value, reflecting strong Brazilian consumer demand for personal care and wellness products.
- Watches & Gifts ranking second suggests a significant portion of purchases are for the purpose of gifting and occasions.
- Bed, Bath & Table incurs the highest freight costs which indicates that bulky, heavy items exists with lower freight-to-revenue efficiency.

5. Revenue by Category (Top 10)

```
#Total revenue according to each product category
query5 = pd.read_sql("""select p.product_category_name, sum(i.price) as total_revenue
from olist_products_dataset as p
join olist_order_items_dataset as i on p.product_id = i.product_id group by p.product_category_name
order by sum(i.price) DESC
limit 10""", connection)
query5
```

Rank	Product Category	Total Revenue
1	Beauty & Health (beleza_saude)	1,258,681.34
2	Watches & Gifts (relogios_presentes)	1,205,005.68
3	Bed, Bath & Table (cama_mesa_banho)	1,036,988.68
4	Sports & Leisure (esporte_lazer)	988,048.97
5	Computers & Accessories (informatica_acessorios)	911,954.32
6	Furniture & Decor (moveis_decoracao)	729,762.49
7	Cool Stuff	635,290.85
8	Home Utilities (utilidades_domesticas)	632,248.66
9	Automotive (automotivo)	592,720.11
10	Garden & Tools (ferramentas_jardim)	485,256.46

6. Segmentation of Customers

Customers were segmented by purchase frequency into three tiers- New, Returning and Loyal using CTEs in SQL.:

```
#Customer Segmentation
query6 = pd.read_sql("""
    WITH customer_category AS (
        SELECT
            c.customer_unique_id,
            COUNT(o.order_id) AS order_count,
            CASE
                WHEN COUNT(o.order_id) = 1 THEN 'New'
                WHEN COUNT(o.order_id) BETWEEN 2 AND 10 THEN 'Returning'
                ELSE 'Loyal'
            END AS customer_segment
        FROM olist_orders_dataset AS o
        JOIN olist_customers_dataset AS c ON o.customer_id = c.customer_id
        GROUP BY c.customer_unique_id
    )
    SELECT
        customer_segment,
        COUNT(*) AS number_of_customers
    FROM customer_category
    GROUP BY customer_segment""", connection)
query6
```

Segment	Definition	Number of Customers
New	1 order	93,099
Returning	2–10 orders	2,996
Loyal	11+ orders	1

- An overwhelming 96.8% of customers are first-time buyers meaning that the focus should be on customer retention.
- Only 3.1% of customers are Returning, and just one customer qualifies as Loyal which is an indication that the platform has not yet succeeded in converting casual buyers into repeat purchasers.

Power BI Dashboard

The accompanying Power BI dashboard gives an interactive single-page view of the following components:

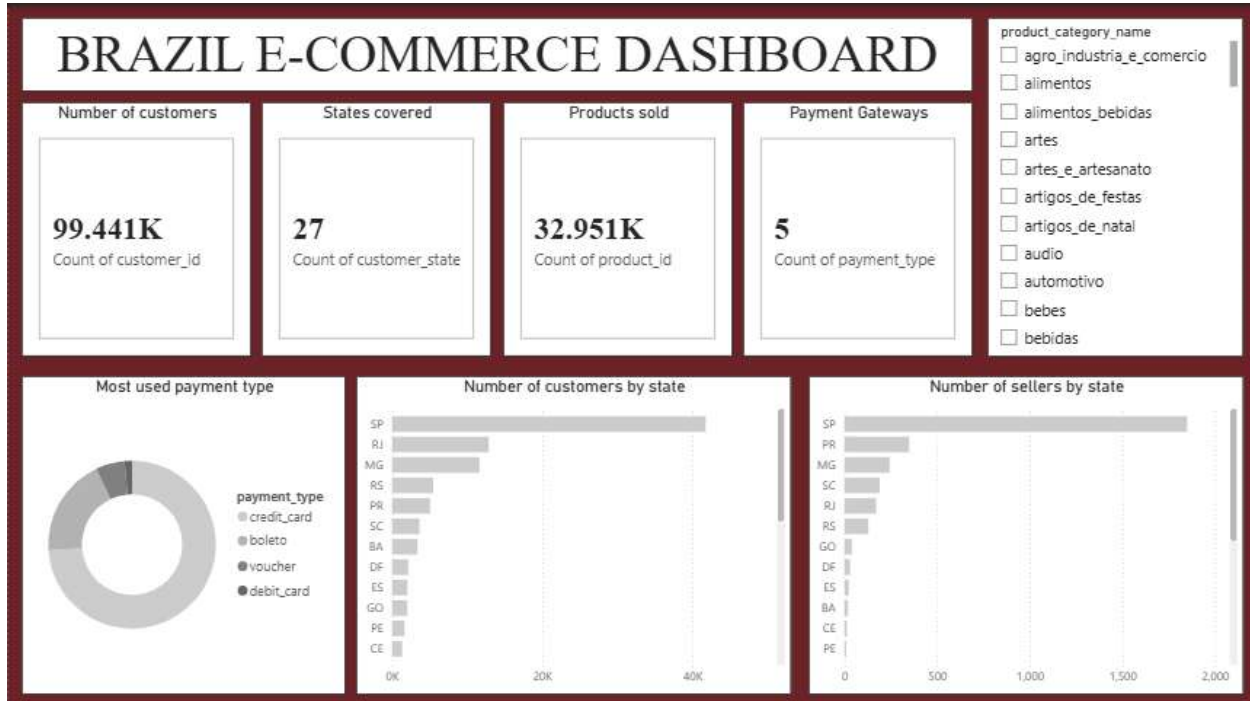
KPI Cards

- Number of Customers: Total unique (each consumer has a unique id) buyers on the platform.
- States Covered: Geographical reach across Brazil.
- Products Sold: Total distinct products transacted.
- Payment Gateways: Count of actively used payment methods.

Charts

- Number of Sellers by State (bar chart)
- Number of Customers by State (bar chart)
- Most Used Payment Type (donut chart)

Slicer for product category enabling a segmented search on the dataset.



Recommendations

Based on the findings above, the following recommendations are proposed:

- **Improve Customer Retention**

- Introduce a loyalty programme that rewards repeat purchases and loyal customers with discounts, early bird sales, etc.
- Personalise product recommendations using purchase history to increase the likelihood of a second transaction.

- **Expand into untapped markets**

- Target campaigns to North-Eastern and Northern Brazilian states where the concentration of orders is low.
- Offer region-specific promotions during local gatherings or cultural events to boost orders.

- **Optimise freight cost efficiency**

- Consider warehousing for cargo to reduce last-mile costs.
- Introduce minimum order thresholds for free shipping in heavy-freight categories to save costs.

- **Invest in product categories that bring more revenue**

- Introduce category-specific panes to convert searches to sales (e.g., 'Top Picks in Bed, Bath and Table').
- Monitor review scores closely for high-revenue categories so as to help customers in case of quality issues.

- **Management of onboarded sellers**

- Create a seller performance dashboard that tracks average review score, freight cost, on-time delivery rate, and revenue contribution.
 - Set minimum performance thresholds (e.g., average review score above 3.5) and provide underperforming sellers with coaching resources.
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